An Introduction to 3D Printing

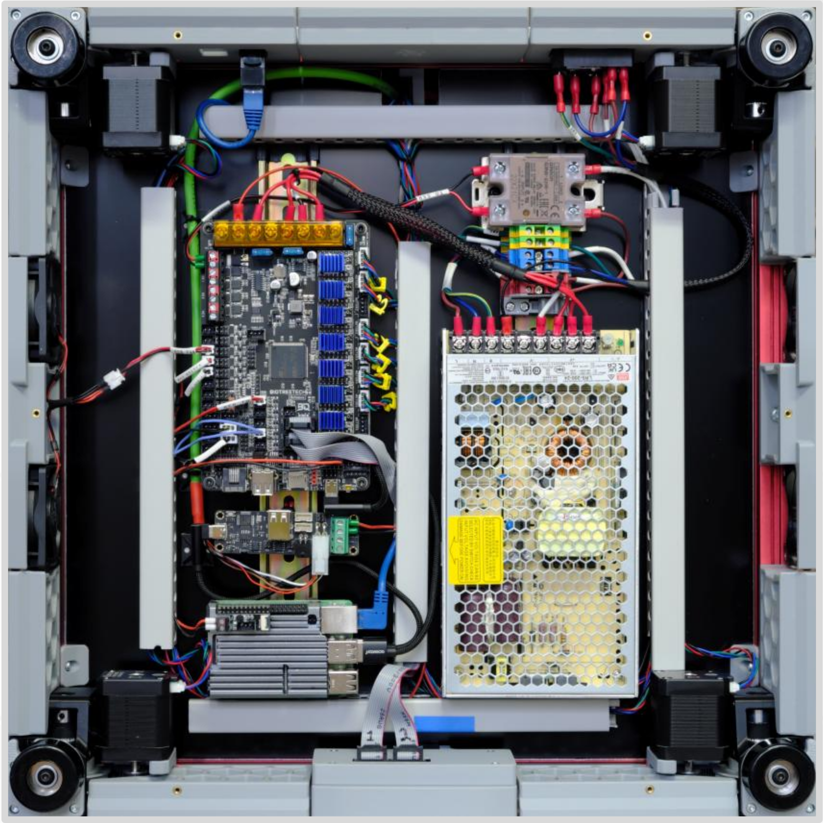
Part 1: FFF Printers

Introduction, Process, Limitations, and Advantages

About Me

- Woodworking and furniture making.
- Built my own printer to learn about 3D printing.
- 6k hours (250 days) of print time.
- Printables.com Flying Gyroscope

(Voron 2.4 GitHub)



FAQ: Which Printer?

Which printer should I get?

- Is the printer the hobby, or is making things the hobby?
- Get the most **reliable** printer you can afford.
  - Do not get sucked into marketing.
  - Do not obsess over spec sheets or speeds.
- Top brands give you excellent slicer profiles out of the box.
  - Very highly recommended for beginners.

FAQ: Tuning

How do I tune my printer?

- Keep it simple to help diagnose problems.
  - Practice restraint when changing slicer settings.
  - Keep the printer stock, unmodified.
- Make one change and try A vs B test.
  - Can you actually see or measure a difference?
- Ellis' Print Tuning Guide.
- OrcaSlicer Calibration Walkthrough.

Outline

Part 1: Printers

Process

Limitations

Dimensional Accuracy

Advantages

Part 2: Practical Application

Examples

Critical Thinking

Design

Part 3: Science

Vibrations and Waves

Linear Motion

Thermal Expansion

3D Printing is a Process

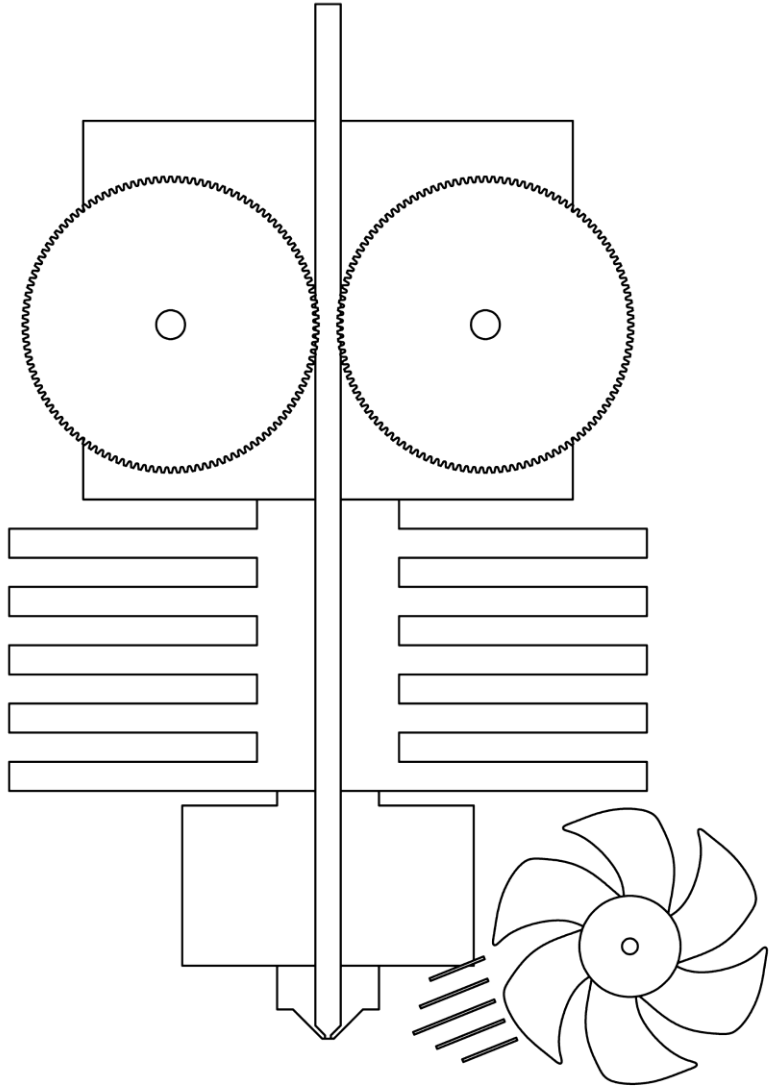
1) Start with a 3D design (digital file).
   - If you download, check the license for restrictions.
   - **STL**. Approximates with tiny triangles.
   - **STEP**. Real geometry. Great for 3D modeling.
   - **3MF**. Container, like a zip file.
     Might contain multiple objects, slicer settings, multi-color settings, …
2) Slice the design.
   - Printers do not accept 3D models.
   - Use software to “slice” objects into vertically-stacked layers.
   - Translate shapes into to Gcode. Standardized commands.

3D Printing is a Process

3) Print. Do you need to …
   - Modify the design?
   - Tune printer settings?
4) Post-process.
   - Remove supports.
   - Assemble.
   - Glue.
   - Fill and sand to remove layer lines.
   - Paint.

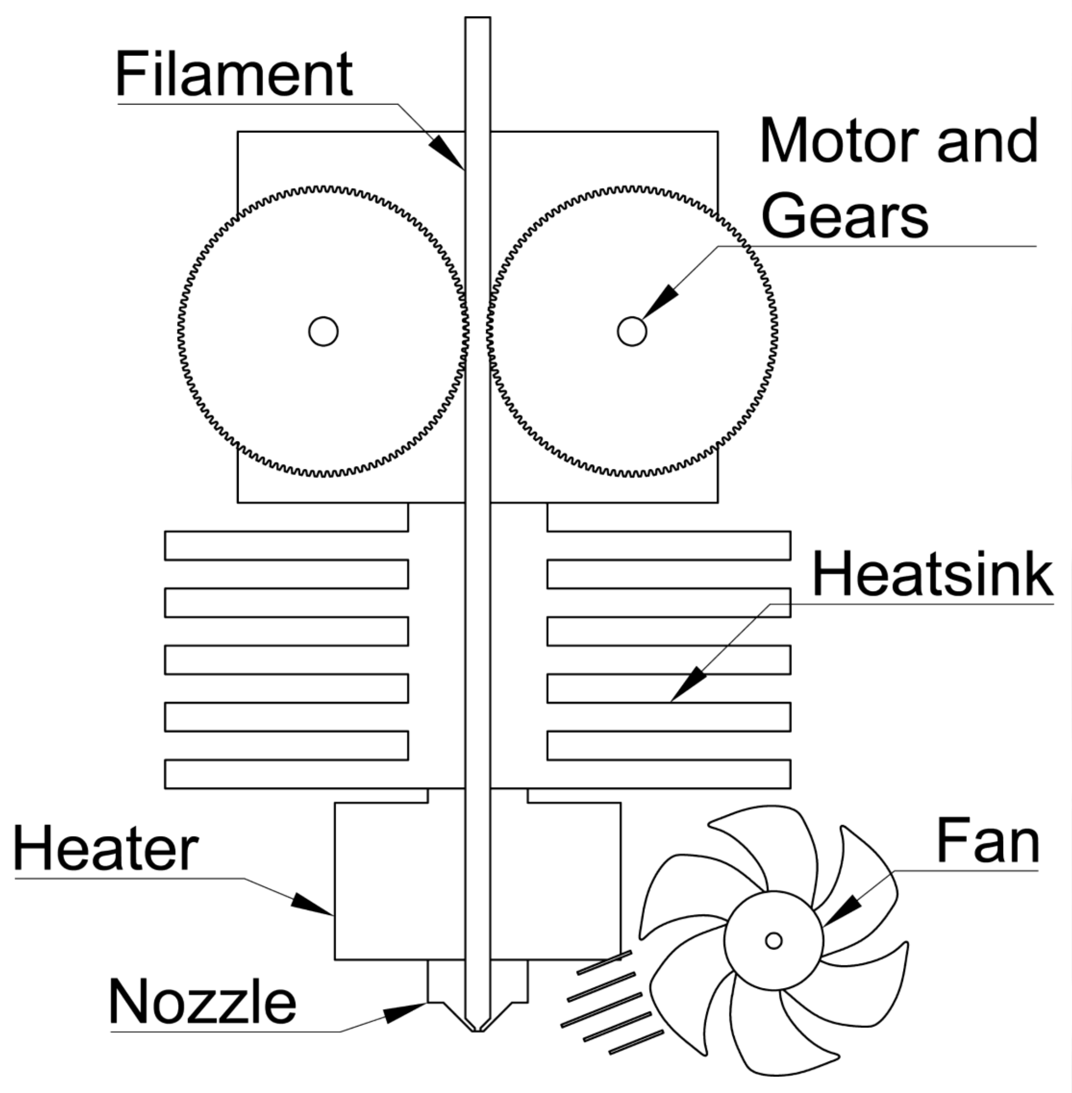
Pushing Plastic

- What kind of 3D printing?
- **FFF. Fused filament fabrication.**
- FDM™. Fused deposition modeling.
- Filament and extrusion-based additive manufacturing.
- Toolhead. Moves around and deposits plastic.



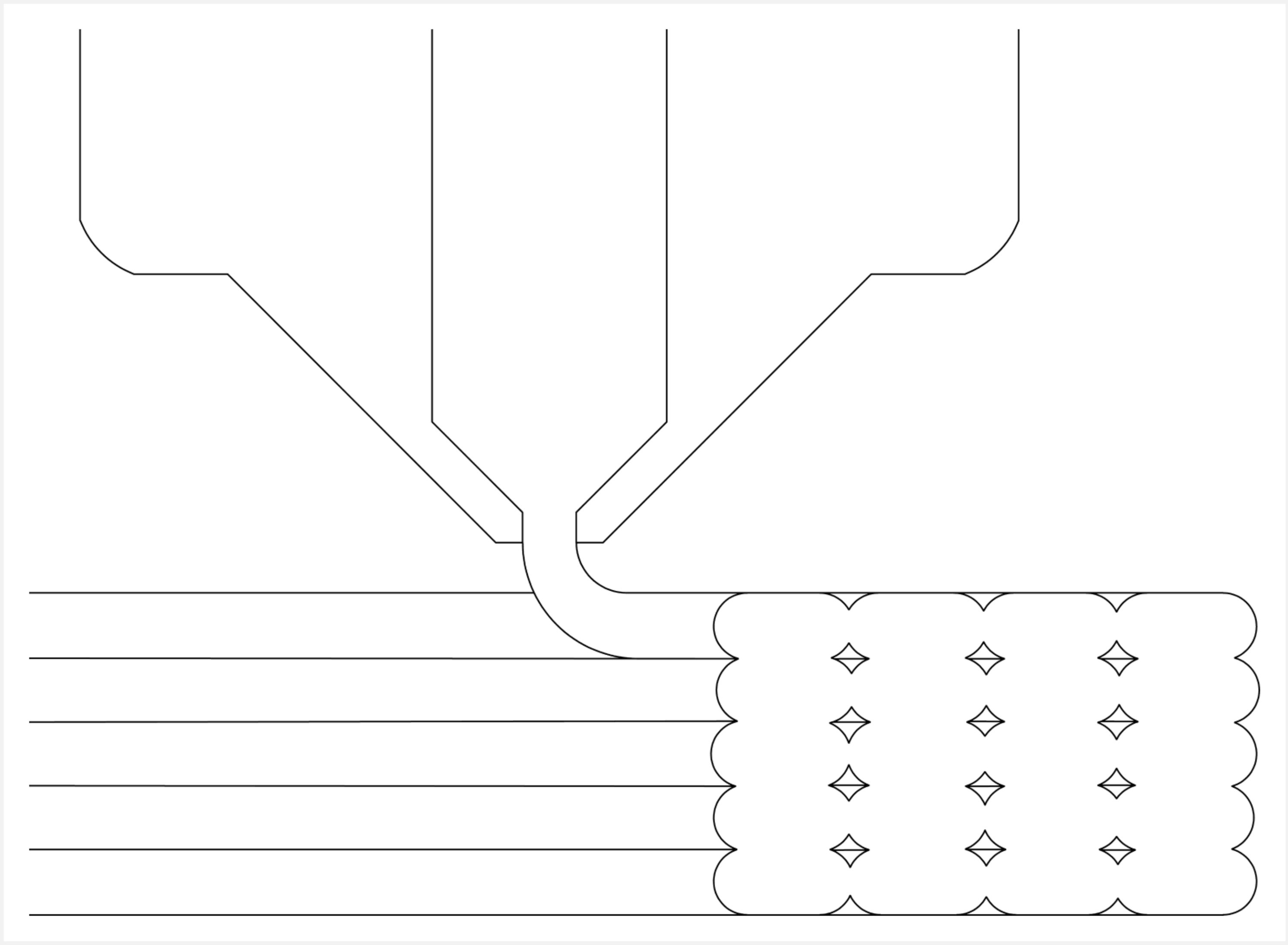
Pushing Plastic

- Filament. Long, continuous plastic noodle.
- Motor and gears push filament.
- Heatsink keeps heat in the heater and nozzle.
- Heater softens filament.
- Nozzle opening determines line width.
- Fan cools hot plastic.



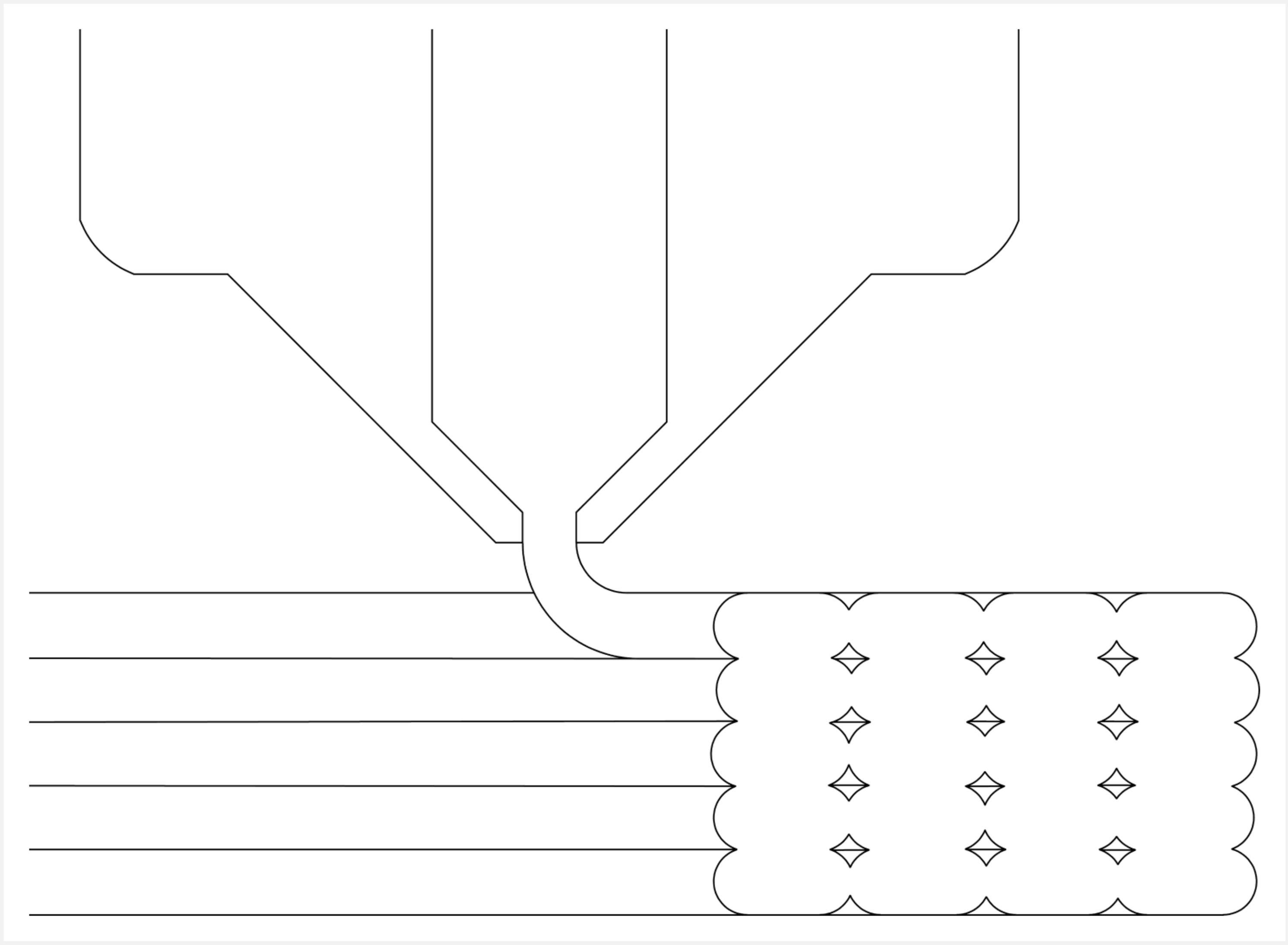
Extrusions

- Toolhead deposits a line called an **extrusion**.
- Best to squish extrusions into place.
- Extrusions lock in place when they cool.
- Printing needs fast cold-hot-cold thermal cycle.



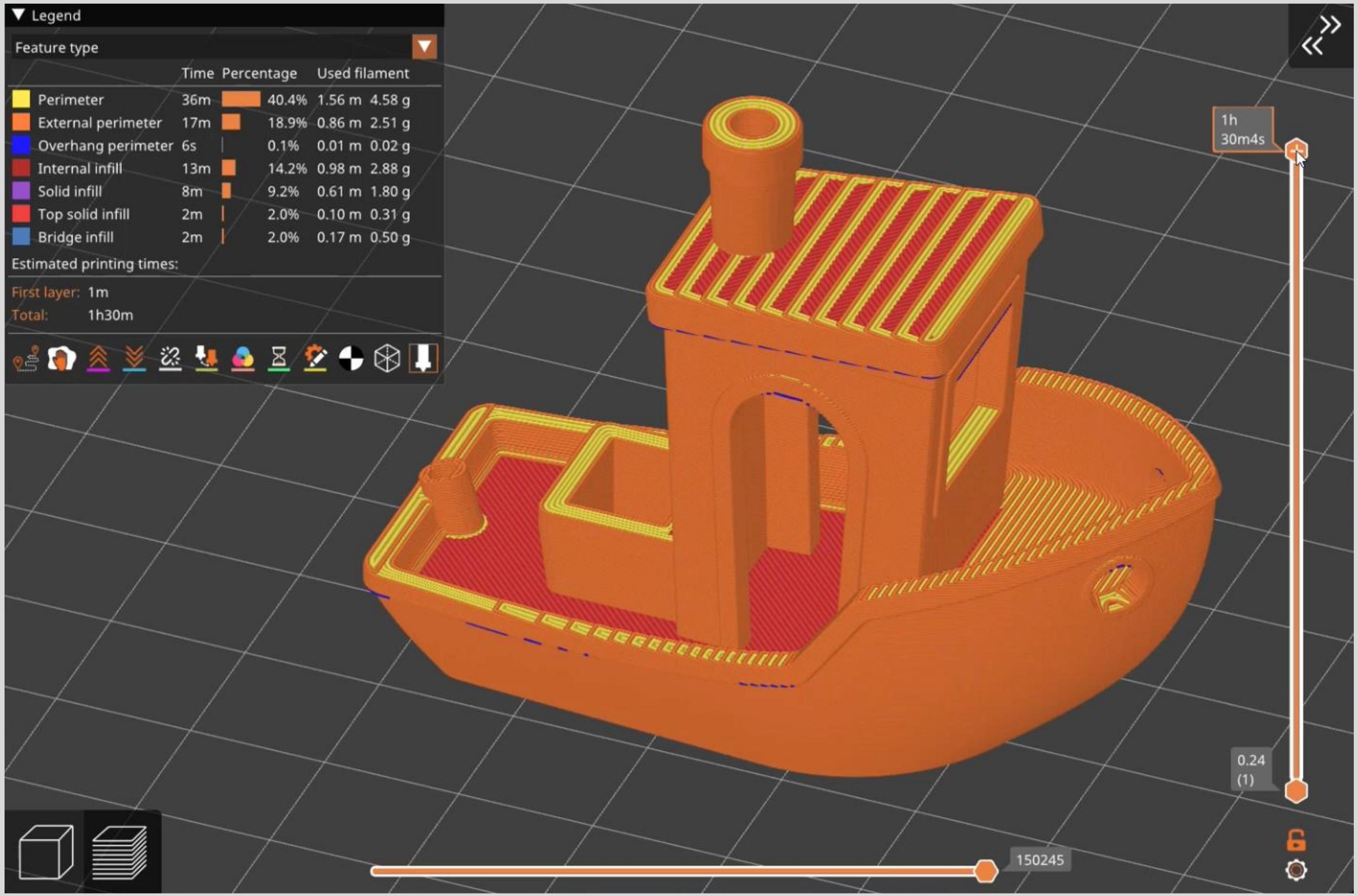
Stacking Layers

- Printers *simulate* 3D shapes by stacking layers.
- You could call it 2.5D printing.



Slicer Layers

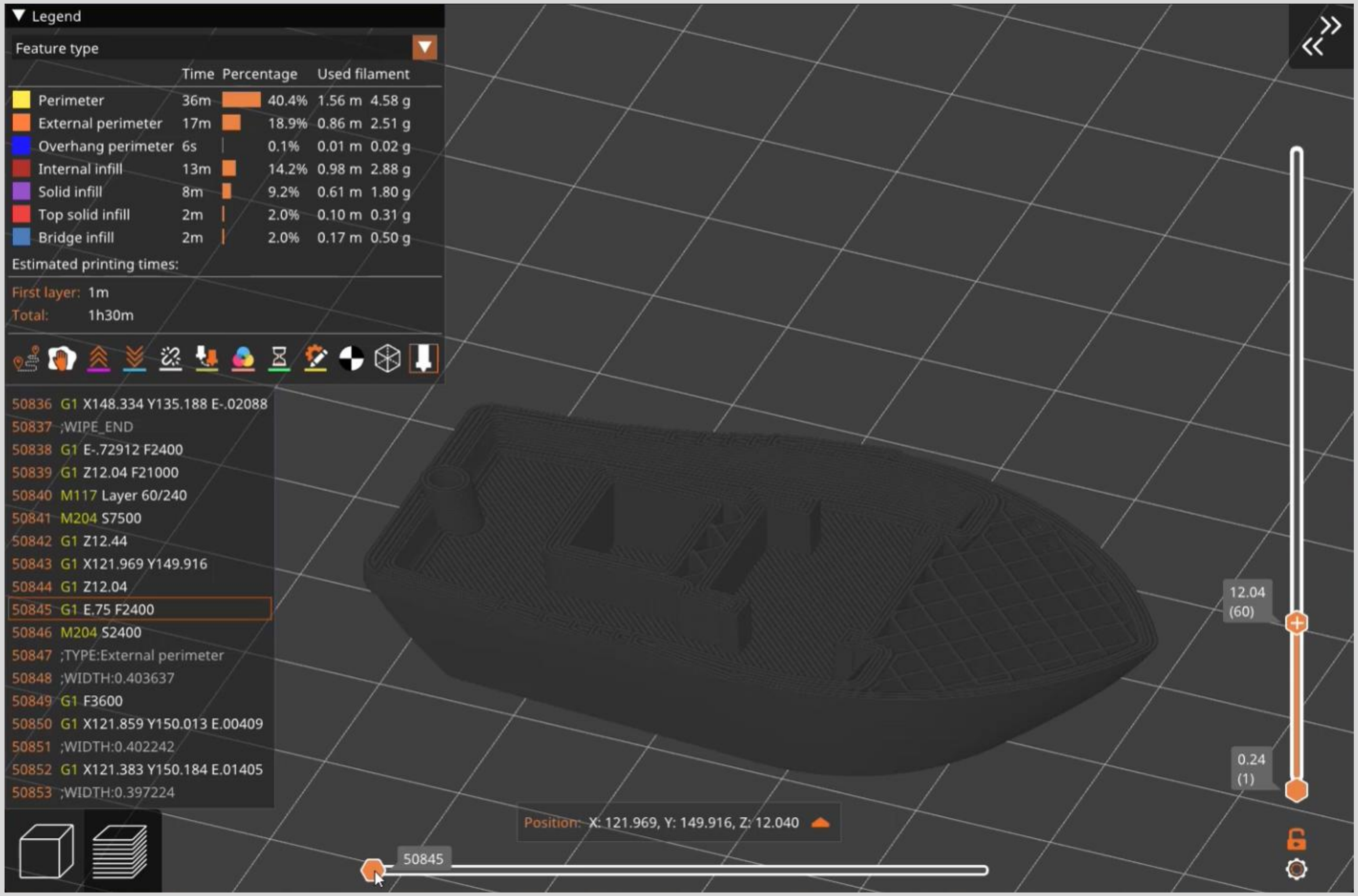
Let's see how PrusaSlicer handles layers.



(Animation, FlyingGyroscope GitHub)

Slicer Layers

- Single layer preview.
- Gcode shows what the printer is actually doing (lower left).
- *G1* = move. Can include X, Y, or Z coordinates.
- *G1 X Y E* = move and extrude.



(Animation, FlyingGyroscope GitHub)

Slicer Layers

- Slicers sort and categorize extrusions.
- Adjust settings for each category. Typically …
  - Printing speed.
  - Cooling (fan speed).

Legend

Feature type

	Time	Percentage	Used filament
Perimeter	36m	40.4%	1.56 m 4.58 g
External perimeter	17m	18.9%	0.86 m 2.51 g
Overhang perimeter	6s	0.1%	0.01 m 0.02 g
Internal infill	13m	14.2%	0.98 m 2.88 g
Solid infill	8m	9.2%	0.61 m 1.80 g
Top solid infill	2m	2.0%	0.10 m 0.31 g
Bridge infill	2m	2.0%	0.17 m 0.50 g

Estimated printing times:

First layer: 1m

Total: 1h30m

(Animation, Thermal camera video by NeedItMakeIt)

Thermal Cycle

- Melting filament is technically incorrect. Why?
    - No transition into a liquid.
- Filament is **softened** into a viscous state.
    - Like a gel.
- Glass transition temperature instead of melting temperature.

Easy Filaments

- **PLA** (derived from corn starch).
  - General-purpose filament for indoor applications.
  - Excellent bridging.
  - Very stiff.
  - Deforms permanently under load (creep).
  - Low temperature resistance — will not survive inside a car on a hot summer day.
- **PETG** (plastic water bottles).
  - Pretty good general-purpose filament.
  - Hygroscopic (needs low-humidity storage).
  - Excellent choice for springs (elastic).
  - Moderate temperature resistance — can be used outdoors.

Easy Filaments

- What makes PLA and PETG easy?
  - Low warp (low shrink).
  - High dimensional accuracy.
  - Room temperature print chamber.

Proper Protocol

Let’s review some everyday guidelines.

- Secure the loose end of a spool of filament **at all times**.
  - Tangles ruin prints and spools, and waste time.
- Do not reach into the printer while it is running.
  - Use the touchscreen or buttons to pause and resume prints.
- Watch the first layer before you walk away.
- Avoid burns.
  - Read nozzle and print bed temperatures on the display.
  - Never reach near a hot nozzle (210 °C = 410 °F).
- Do not leave fingerprints on print sheets.

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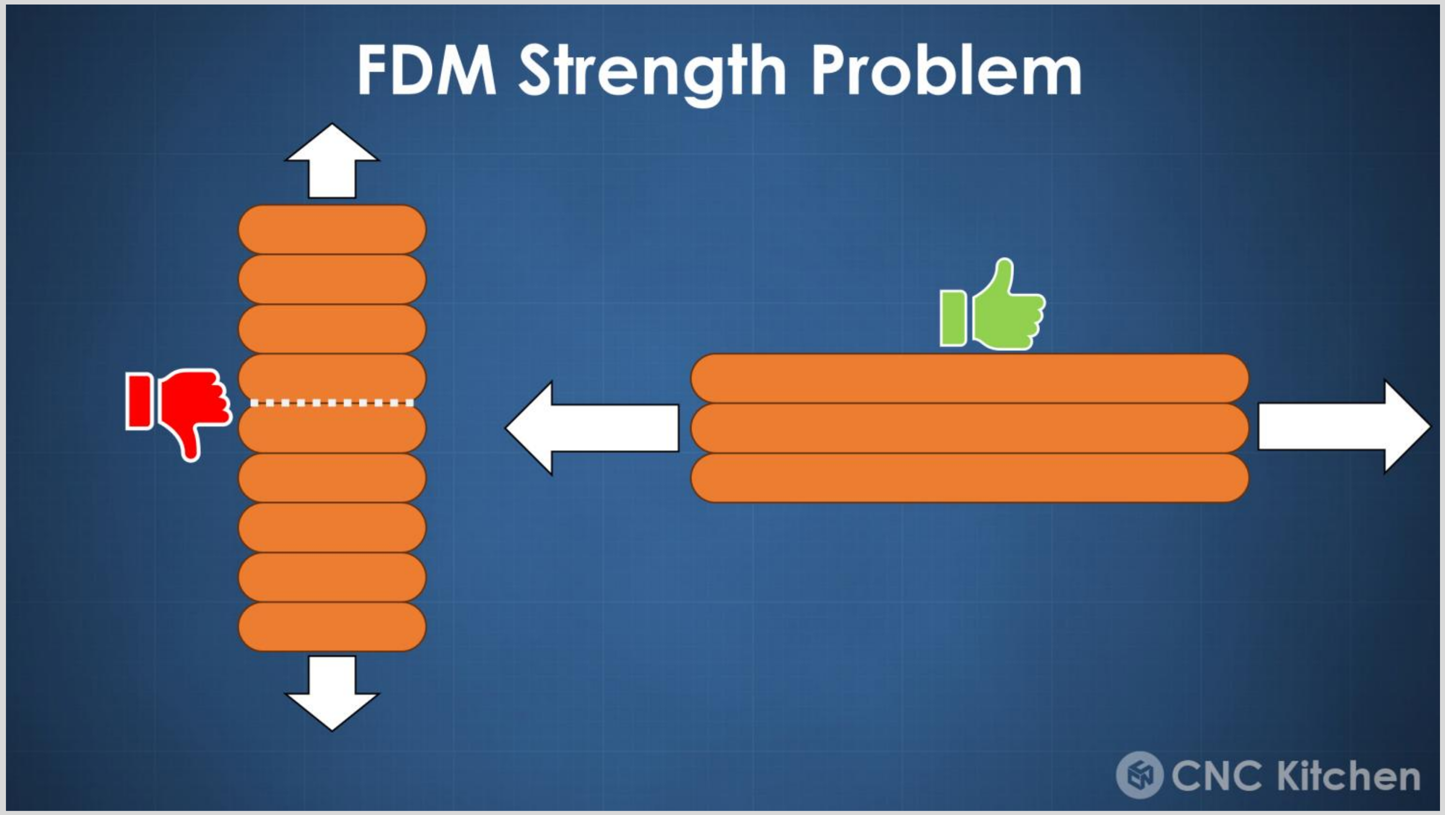
Thermal Cycle

- Some thermoplastics shrink as they cool.
  - Up to 1 or 2%.
  - Seems small, but it can distort and skew dimensions.
  - Cracking and delamination in extreme cases.
- Advice?
  - Tightly control ambient temperature and cooling rate.
  - Some filaments (ABS, Nylon) like it hot → extra heating.
  - Some filaments (PLA) like it cold → extra cooling.
  - PETG is somewhere in-between.

Anisotropic Strength

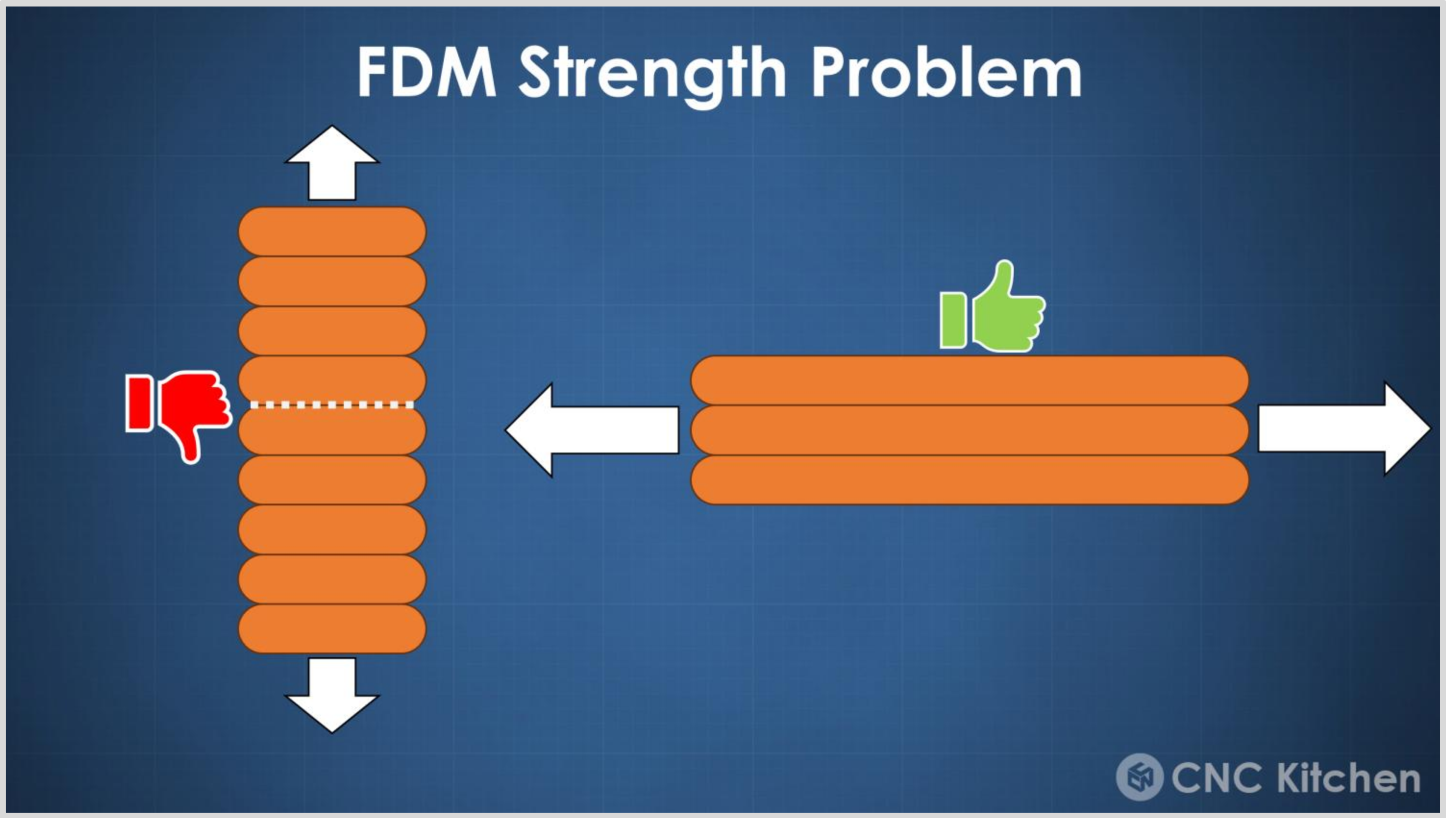
- Print orientation and layer lines affect strength.
- Layer-to-layer bonding is a weak point!
- Within an extrusion, hot filament bonds to hot filament.
- For a new layer, hot filament bonds to a warm layer.



(Picture from CNC Kitchen)

Anisotropic Strength

- The “wrong” orientation can lose 50% of strength.
- Advice?
- Avoid tall and skinny shapes.
- Improve layer bonding.
    - Increase printing temps.
    - Decrease printing speed.



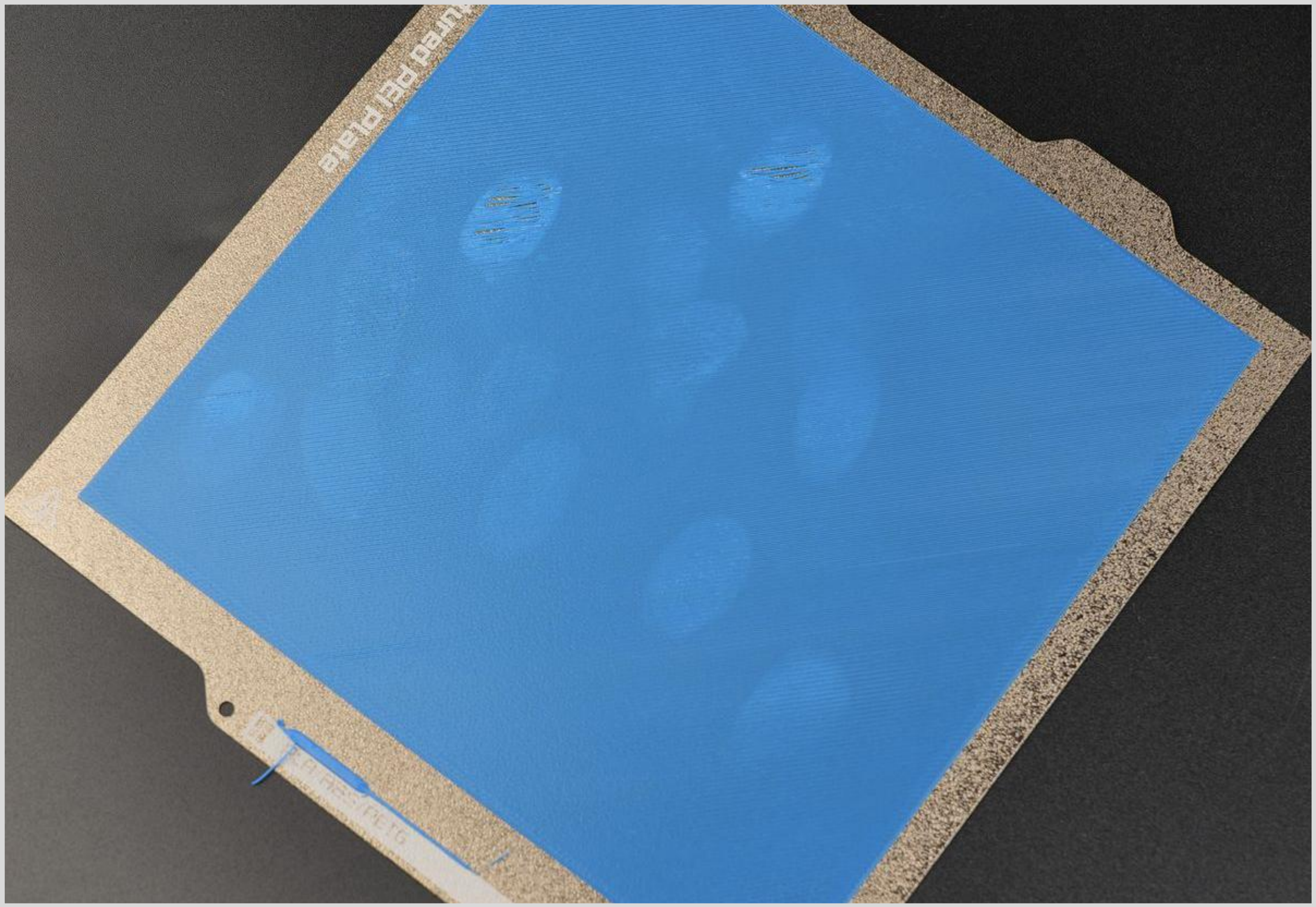
(Picture from CNC Kitchen)

First Layer

- What is the most important layer in a 3D print?
  - The first layer!
  - First layers keep everything locked in place.
- Advice?
  - **Always watch the first layer.**
  - Tune first layer squish.
  - Flat surface area on the bottom of the 3D object.
  - Heated print bed.
  - Perfectly clean print bed.
  - Apply adhesion promoters (glue).

First Layer Mistakes

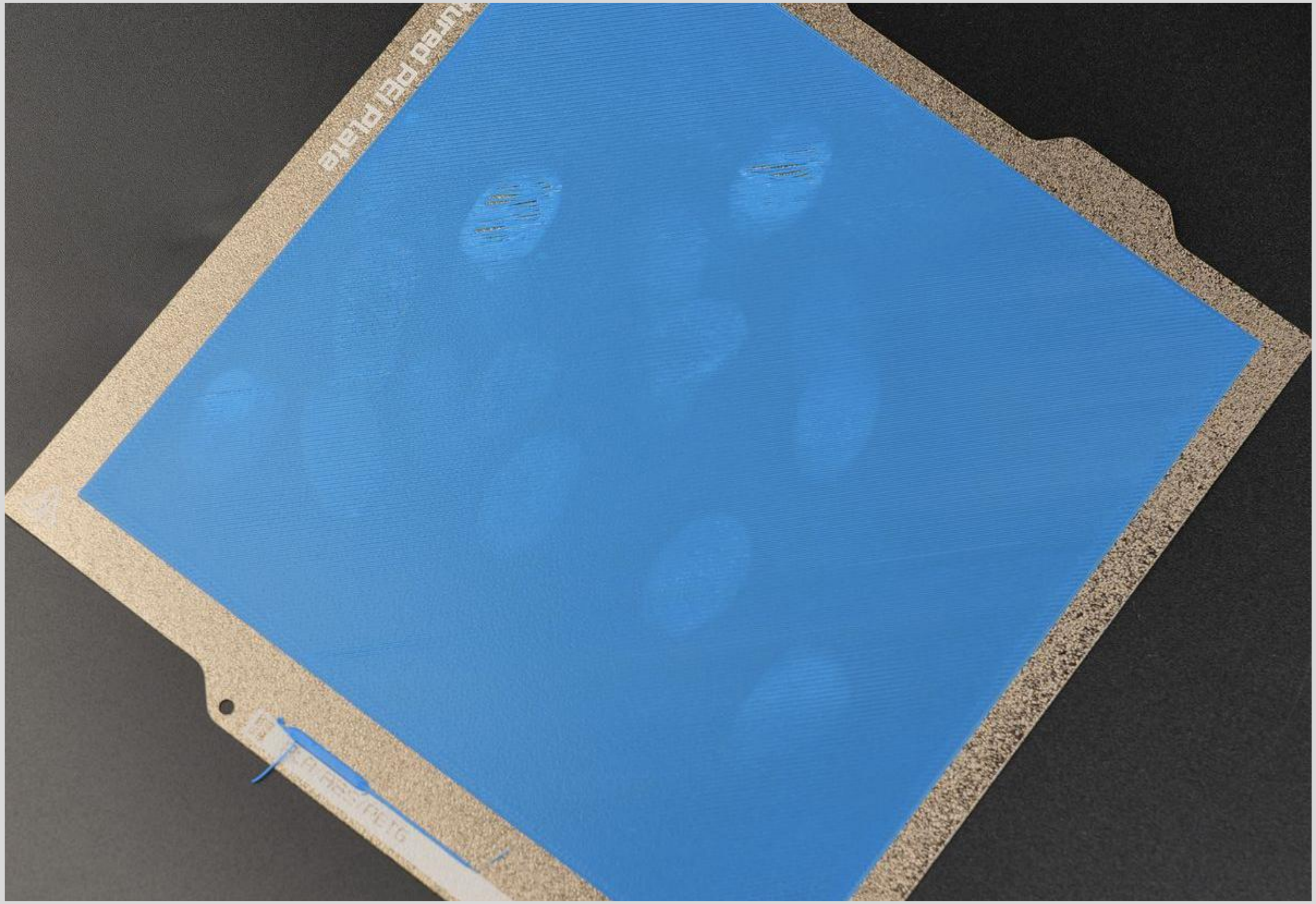
- What is wrong with this picture?
- **Fingerprints!** Plastic does not stick to oily fingerprints.



(Picture from Bambu Labs Wiki)

First Layer Mistakes

- Advice?
- Deep cleaning: dish soap, soft brush, and water.
- Everyday use: window cleaner with ammonia.



(Picture from Bambu Labs Wiki)

First Layer Mistakes

- What happens when a first layer does not stick?
- The print breaks free and makes “spaghetti”.
- You can see when things went wrong.
- Some of the rectangular print finished.

(Screenshot from Everything STEM)

Stair-Stepping

- Printing with layers → Vertical slopes become stair steps.
- Mostly cosmetic effect.
- Advice?
- Rotate objects to hide layer lines.
- Smaller layer height. Increases print time.

(Picture from Hamburg University Research Project)

First Layer Warping

- Why do corners and edges peel up?
- Temperature gradient from top to bottom.
- Regions shrink at different rates →
  Internal stresses →
  Stresses concentrate in corners.
- Shrinkage is a percent →
  Large parts have the most trouble.

(Picture from Creality Cloud Blog)

First Layer Warping

Consider this analogy.

- Imagine you want to remove a label or piece of tape.
- What do you do?
- Start peeling up the corner because that is the weak point.
- Same thing here.
- Physics found the weak point.

(Picture from Creality Cloud Blog)

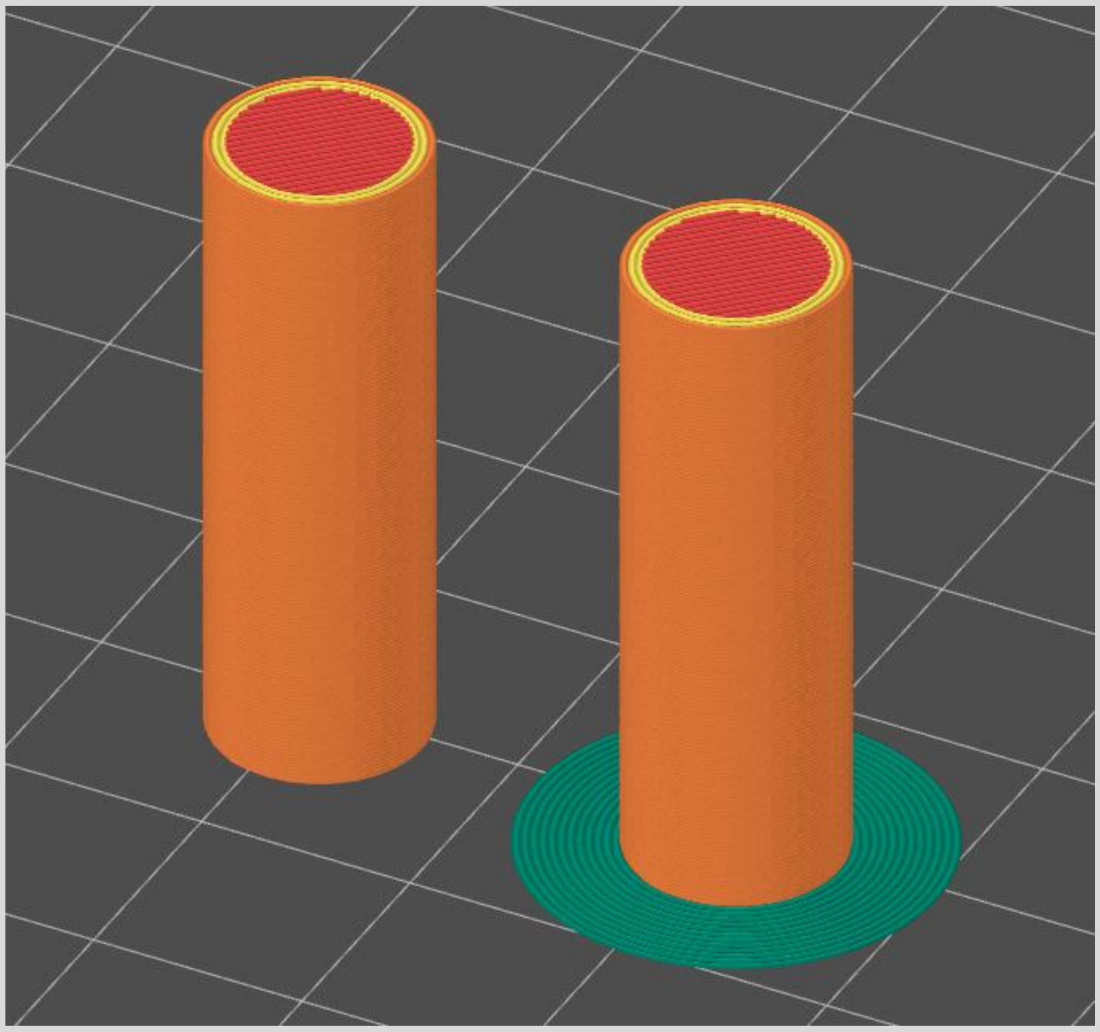
First Layer Warping

- Advice?
- Better control over temperatures.
  - Filament temperature.
  - Print chamber temperature.
  - Cooling fans.
- Evenly distributed cooling.
  - Do not want fans or drafts that cool one side.
- Round corners to reduce stress concentration.

(Picture from Creality Cloud Blog)

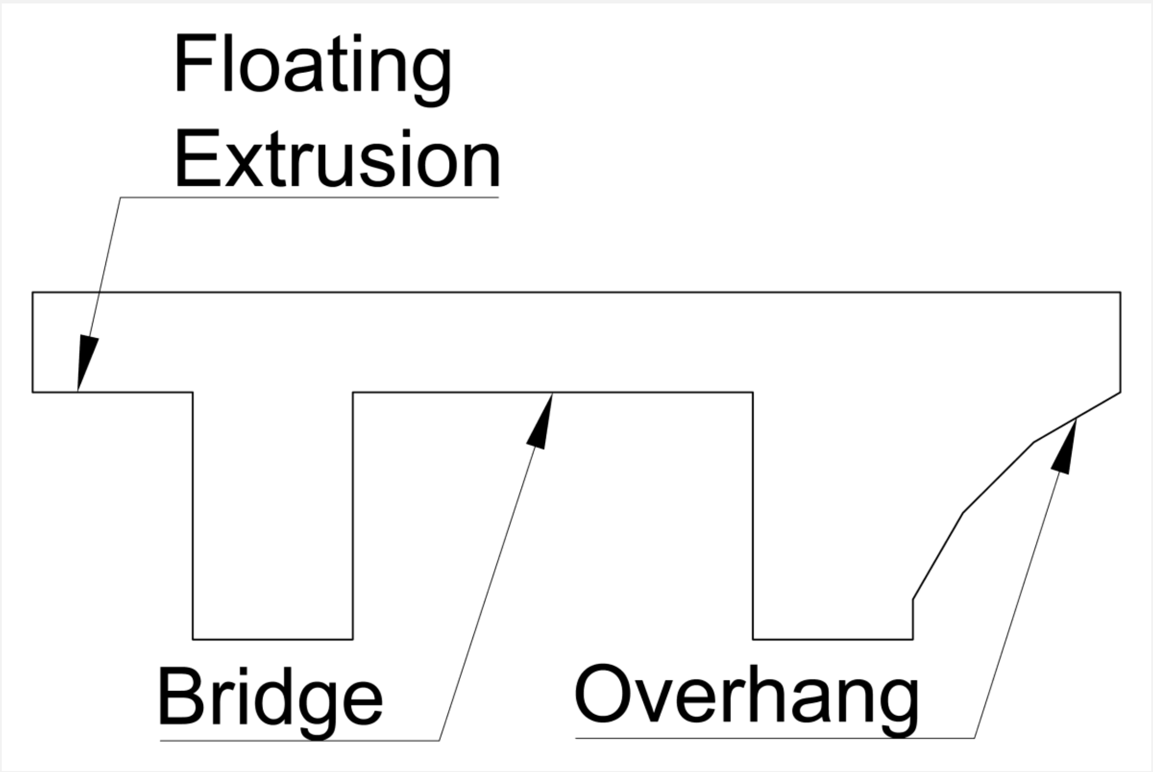
Brim

- Fights against first layer warping with extra bottom surface area.
- Concentric extrusions joined to bottom perimeter. (Green lines on the right.)
- Typically added with a slicer.
- One layer tall.
- Easily peeled away after printing.



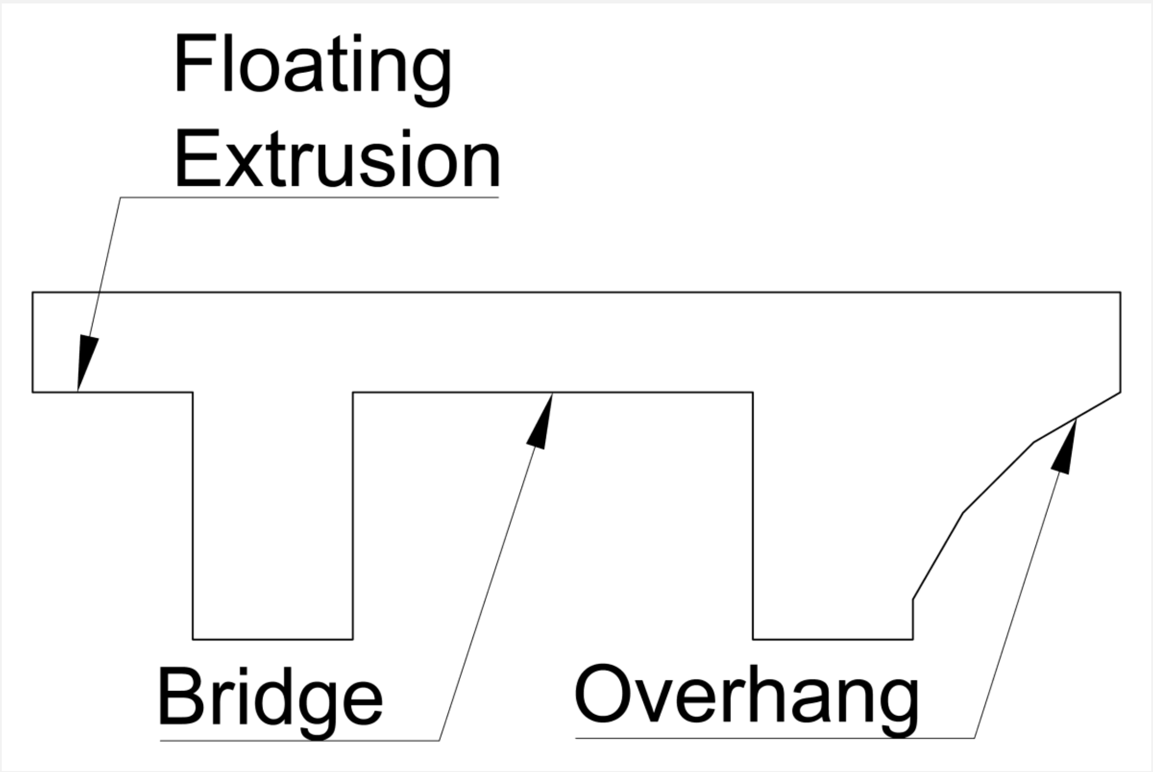
Levitating Layers

- *Critical thinking:* What happens when you try to levitate extrusions?
- Extrusions work best when you press them onto a solid surface.
- Predict what this print will look like.
- What parts will be the most deformed?



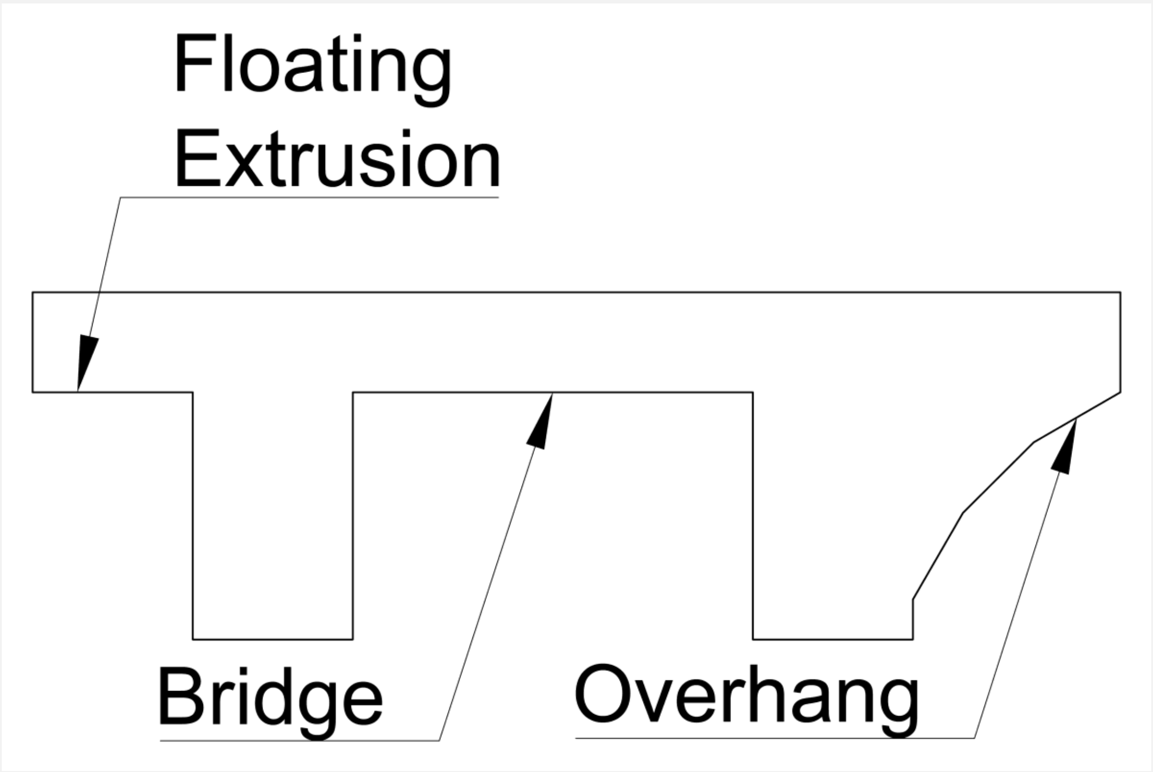
Overhang

- Extrusions that lean over an edge.
- Only the outermost extrusion is affected.
- Partially supported along one side.
- Advice?
- Keep overhangs at 45° to be safe.



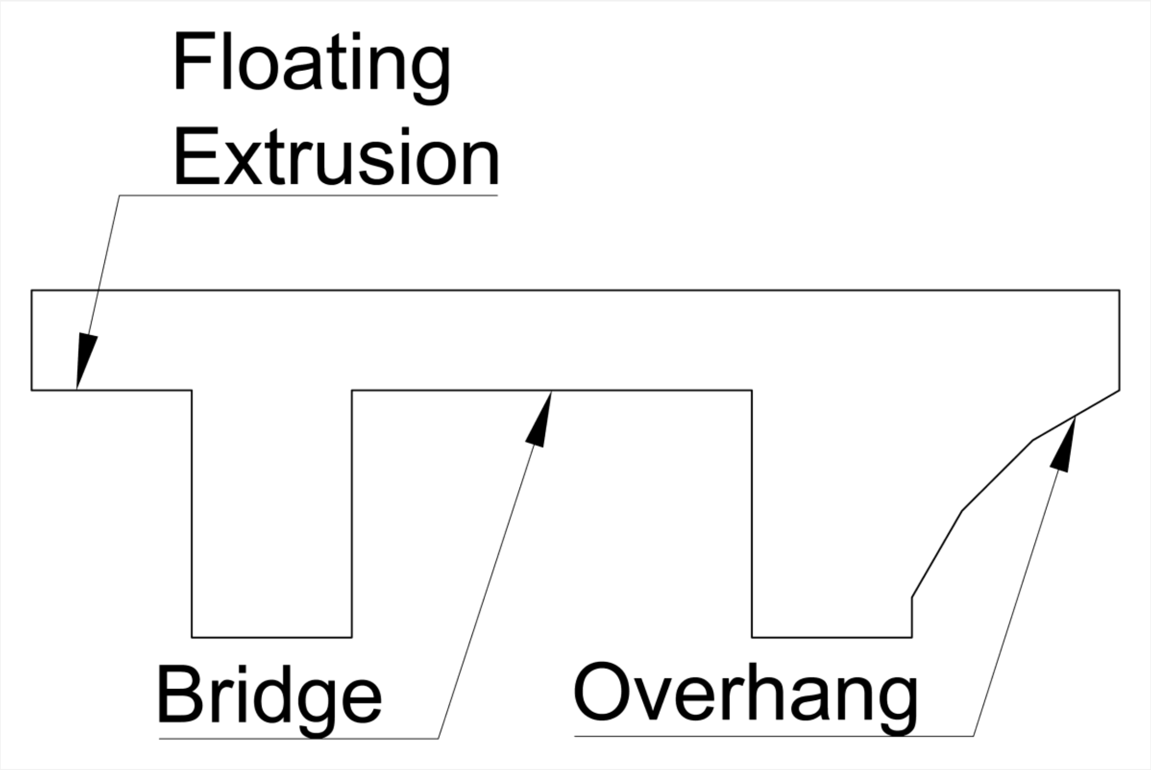
Bridge

- Both ends of an extrusion are supported and anchored.
- Middle hovers over a gap.
- The first few layers sag and droop down.
- Eventually recovers to flat layers.
- Advice?
- Keep bridges as short as possible.



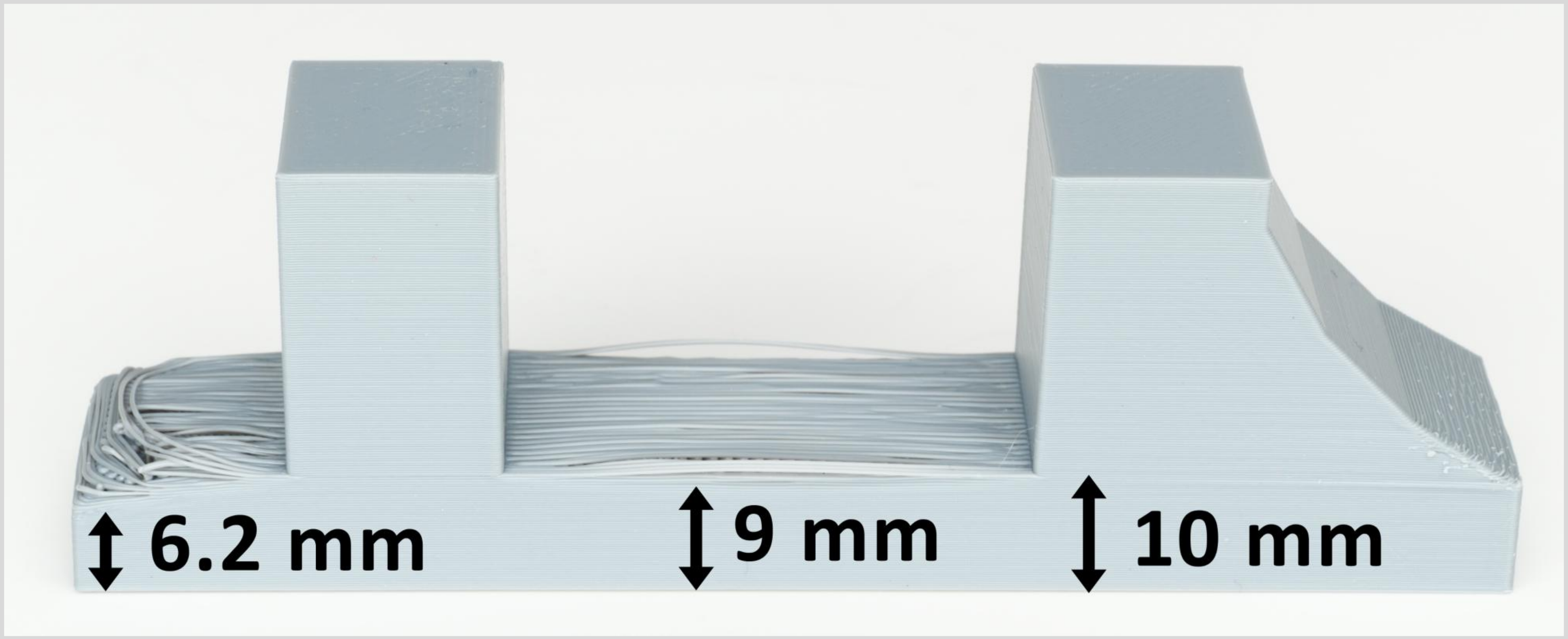
Floating Extrusion

- Extrusions that hang in the air.
- At least one end is not anchored to a solid surface.
- Advice?
- **Avoid. Very unreliable.**



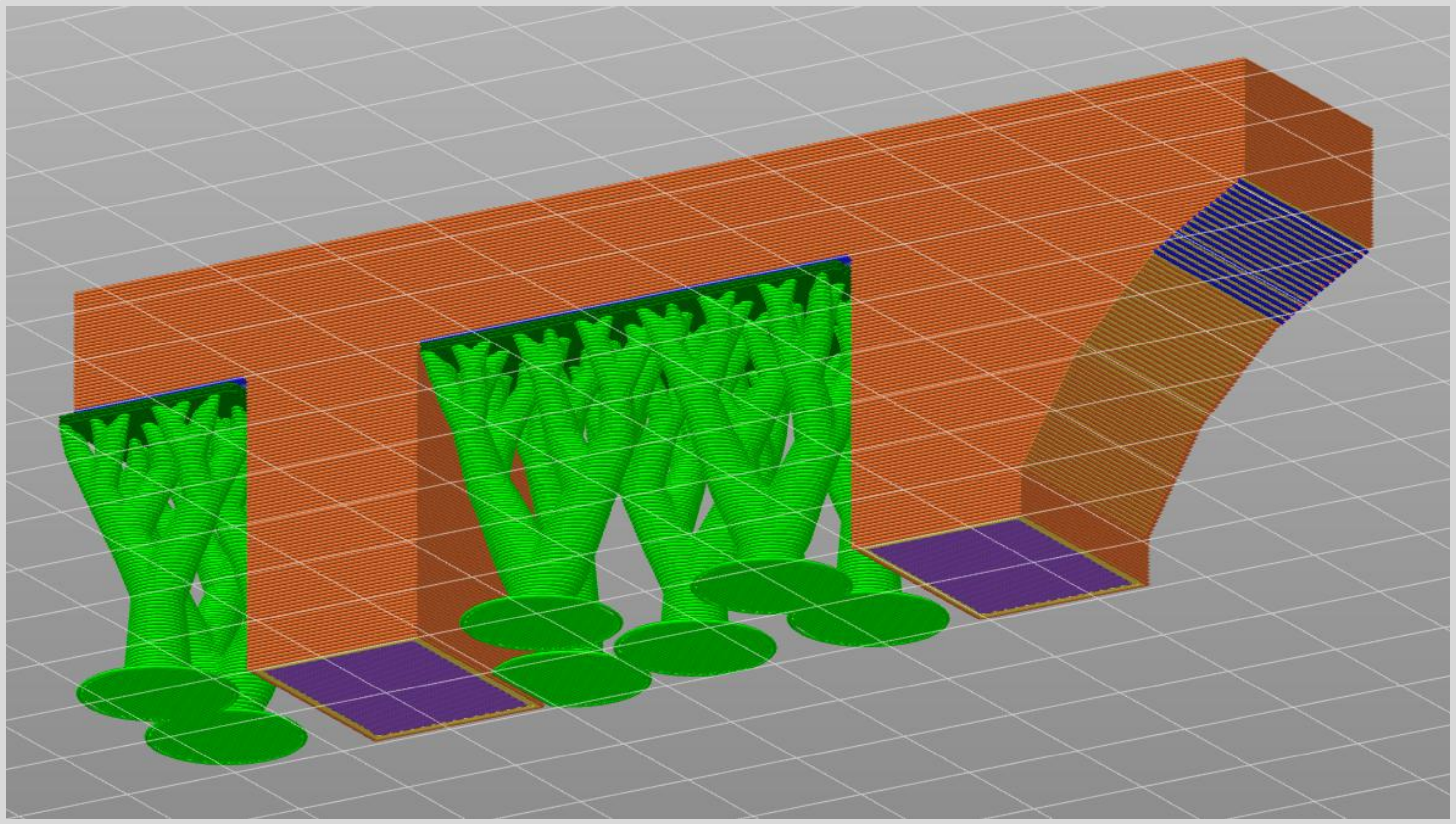
Stress Test Results

- All overhang angles look good.
- Bridge recovers to flat layers in 1 mm (**5 layers**).
- Floating extrusions recover in 3.8 mm (**19 layers**).



Supports

- Supports. Temporary scaffold that fixes:
    - Floating extrusions.
    - Long bridges.
    - Steep overhangs.
- Added by the slicer. (Green trees-like branches.)
- Extra complexity, plastic, and print time.

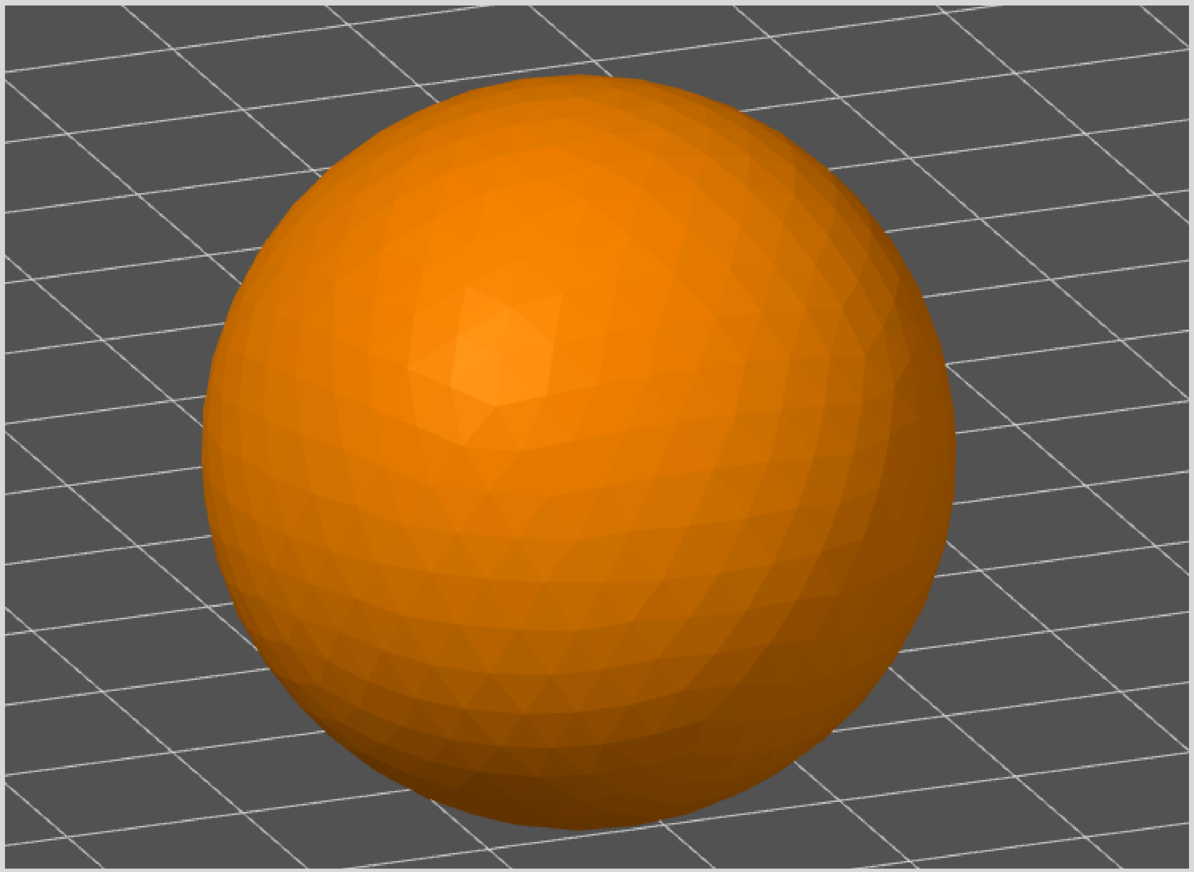


Supports

- In special situations, you can include supports as part of a 3D model.
- Tree/organic supports can reach into difficult spaces.
- Supports are a delicate balance. Why?
  - If they do not bond well enough … they hold nothing in place.
  - If they bond too well … they become part of the print.
- Supports can leave a rough surface finish.
- Sometimes you need to cut away pieces of support.

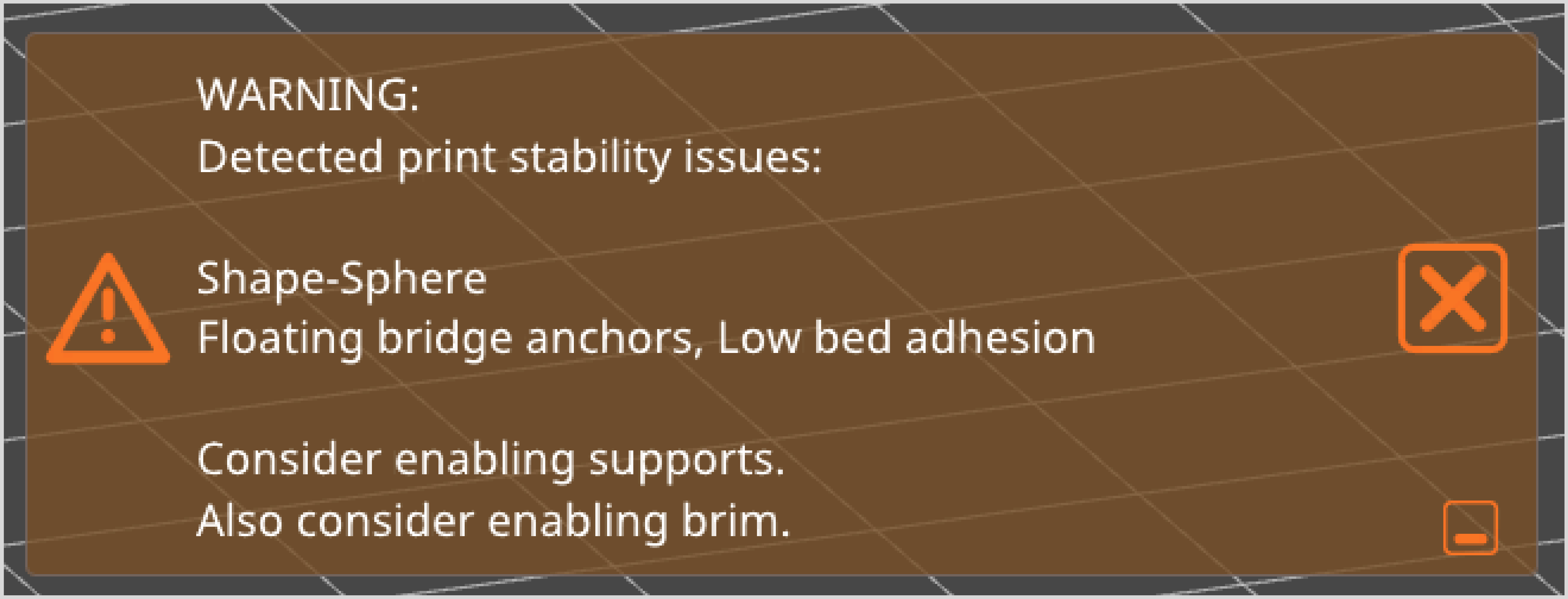
Limitations Review

- Printing a ball runs into some FFF limitations.
  - Very bad **first layer** contact. It will make spaghetti.
  - Steep overhangs on the bottom.
  - Second layer is steep enough to be a floating extrusion.
  - Stair-stepping creates uneven surface finish.



Limitations Review

- PrusaSlicer recognizes these problems.



Limitations Review

How can we fix this print?

- Cannot fix stair-stepping by changing print orientation.
  - Use small layer heights to reduce visual artifacts.
  - Use adaptive layer heights to make the surface look more uniform.
- Brim is marginally helpful.
  - Object starts with too little surface area.
- Add supports for most of the bottom.
- Print two halves, then put them together.

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Dimensional Accuracy

- Do not design parts with exact dimensions.
- This is a trap for beginners.
- Why?
  - Your prints are not exact.
  - All measurements come with **error bars**.
  - The fit is probably too tight to work as intended.
- Where do these errors come from?
- And how do we fix it?

Dimensional Accuracy

- Sources of error include (but are not limited to)
  - Shrinking and warping.
  - Inconsistent extrusions.
  - Misaligned parts.
  - Bent parts from shipping damage.
  - Incorrect belt tension.
  - Backlash, slop, or loose parts.
  - Moving parts that eventually wear down.
  - Unoptimized slicer settings.
  - Thermal expansion (discussed in Part 3).
  - Vibrations (discussed in Part 3).

Modeling Calibration

- *(Printed size) = (Size) × (Scale factor) + (Constant offset).*
- Simplified model.
  - One dimension, like X or Y.
  - Assume errors come from filament and extrusion.
- Scale factor.
  - Error changes with size.
  - Part shrinkage.
- Constant offset.
  - Error is always the same.
  - Extrusion inconsistencies. Over and under-extruding.
  - Surface roughness. Filler or additives like carbon fiber, wet filament.

Scale Factor

- Use scale factor with …
  - Large parts (over 6 inches, 150 mm).
  - Filaments that shrink (like ABS).
- Adjust **XY scaling** or **shrinkage compensation** setting in your slicer.
- Not important for small parts.
- I often ignore XY scale factor in my prints.

Constant Offset

- Very important for any project with assembled pieces.
- **Add clearance (air gap) between parts that touch.**
- Best to add clearance while designing 3D objects.
  - XY offset settings in a slicer are less effective.
- I think about XY constant offsets in all of my designs. Very important!

Adding XY Clearance

- How much XY clearance?
- It depends, but not very much for modern printers.
- *Example:* Making a hole for a bolt.
  - Add 0.2 - 0.3 mm of clearance for a close fit.
- *Example:* Making a lid that easily slides into place.
  - Add 0.4 - 0.8 mm of clearance for a loose fit.

Adding Z Clearance

- How much Z clearance?
- It depends, but less important than XY clearance.
- One layer or less.
  - 0.1 mm.
- Factors that increase clearance.
  - Bridges.
  - High and low spots on print bed.

Recommendations

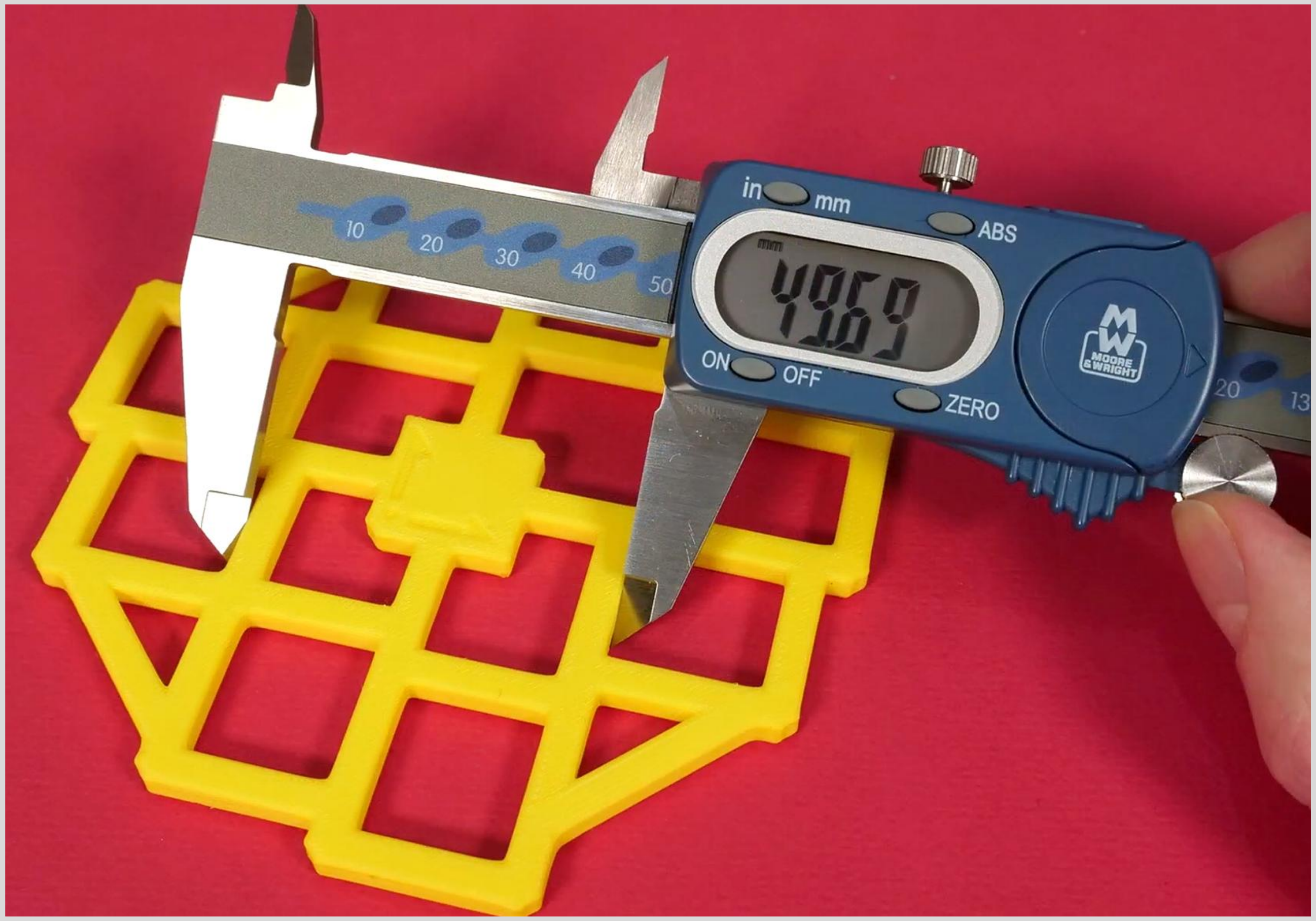
- You sometimes need scale factor.
- You always need constant offset.

Recommendations

- *Caution:* Calibration prints cannot fix everything.
- Calibration cubes.
  - 20 mm cube. Labels for X and Y-axis.
  - Constant offset for outside dimensions.
- Clearance test by 3DMakerNoob.
  - Print-in-place clearance/tolerance.
  - Alters the gap between pieces.
  - See which pieces can move freely.

Recommendations

- Califlower by Vector3D.
  - Inner dimensions.
  - Outer dimensions.
  - Shrinkage.
  - Skew. (If your rectangles turn into parallelograms).



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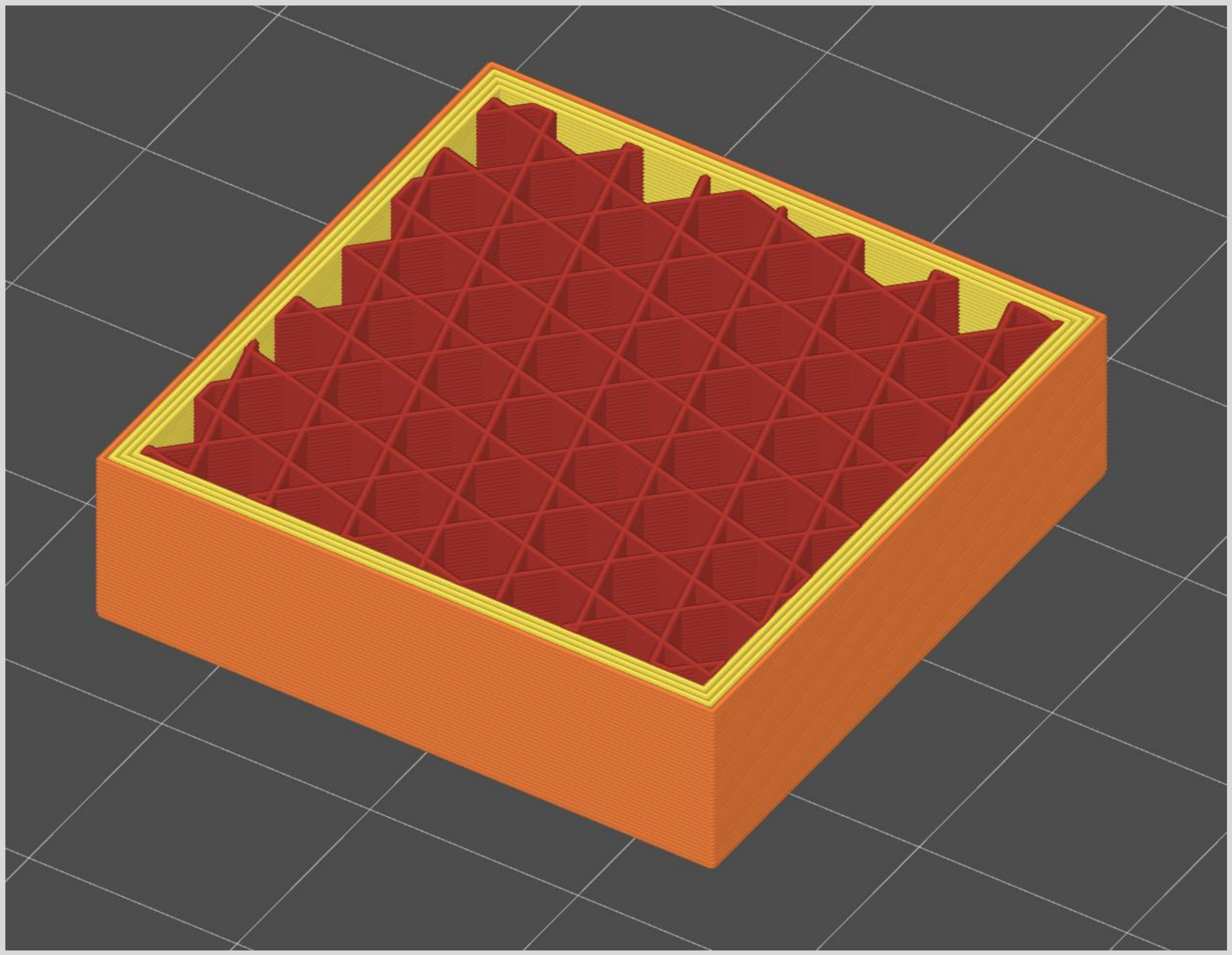
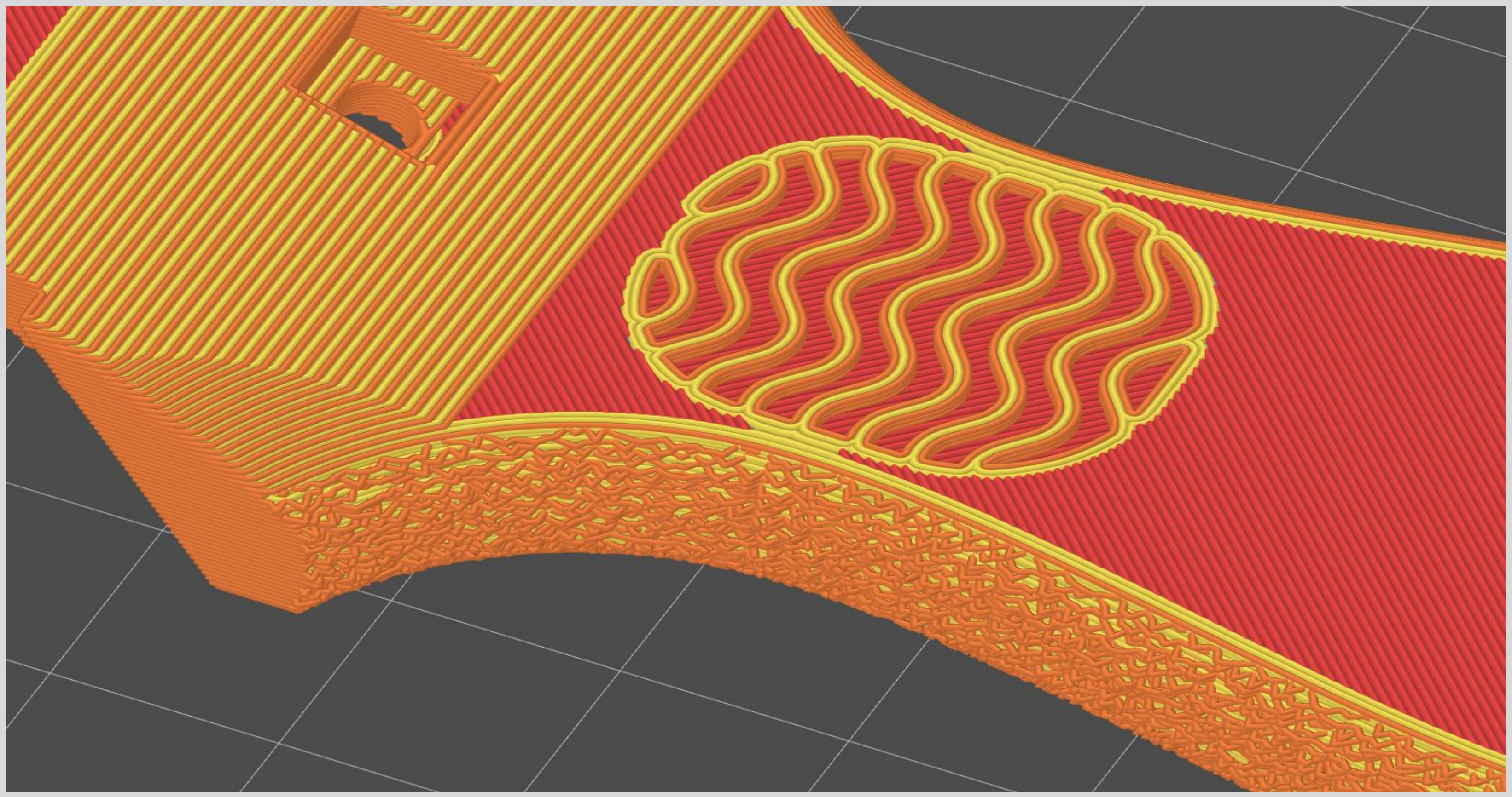
FFF Advantages

- Printers use tiny “building blocks.”
- Typical extrusions.
  - **0.2 mm tall** (1-2 pieces of paper).
  - **0.4 mm wide**.
- Not limited to uniform prints.
  - *Example:* Solid exterior and sparse infill.
- Great at making hollow objects.
  - Imagine making this shape with wood or metal – much harder.



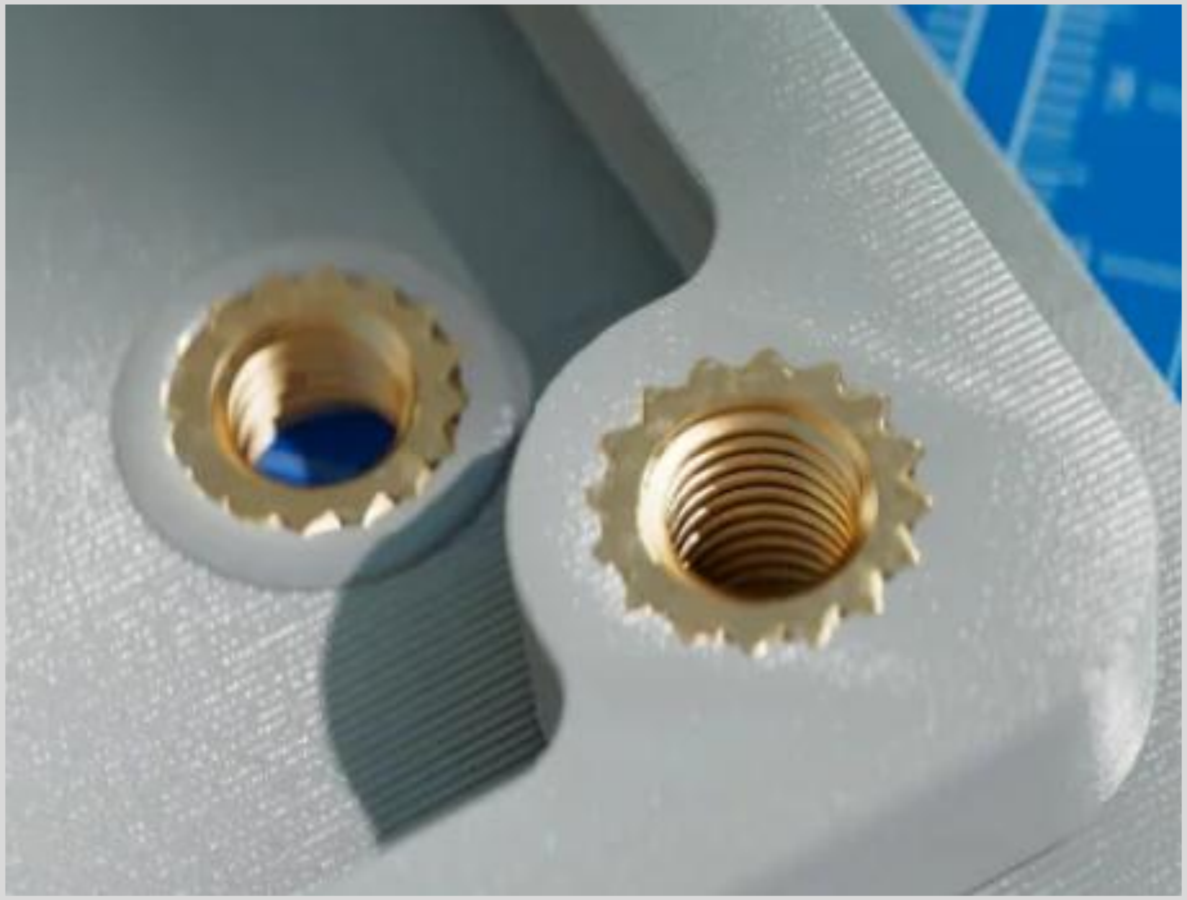
Custom Slicing

- Change slicer settings for specific regions or surfaces.
- *Example:* Add texture to handles with fuzzy skin (bumpy extrusions).



Heat Set Inserts

- Plastic is soft →
  Screws easily wear out plastic holes.
- Heat set insert. Threaded anchor for screws and bolts.
- Install with a hot soldering iron.
- Locks in place when it cools.
- Very durable.



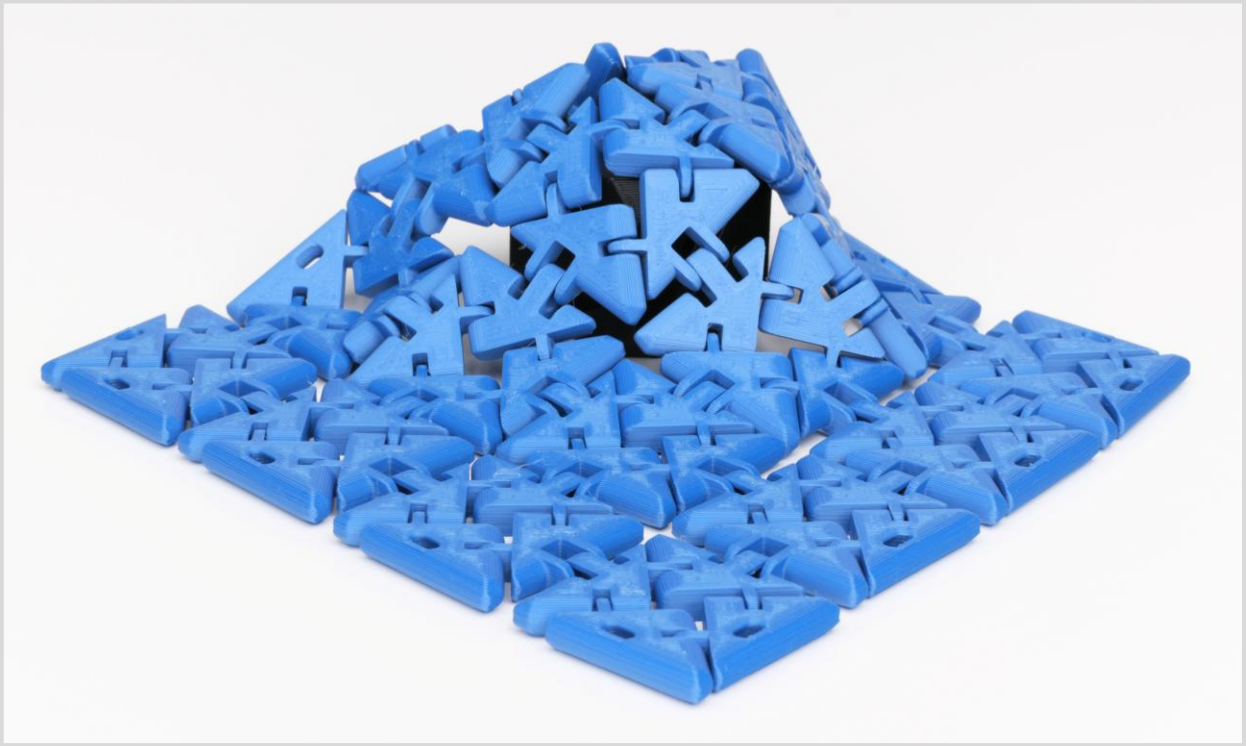
(Pictures from CNC Kitchen)

Captive Parts

- Insert an object into a pocket or void in the middle of a print.
  - Nuts and fasteners.
  - Magnets.
  - Weights.
  - Reinforcing plate
- Object is locked inside when the print finished.
- Add a pause command in the Gcode.
- Object must sit below the current layer, or it creates a collision with the toolhead.
- Waiting around for the right layer is inconvenient.

Print-in-Place (PIP)

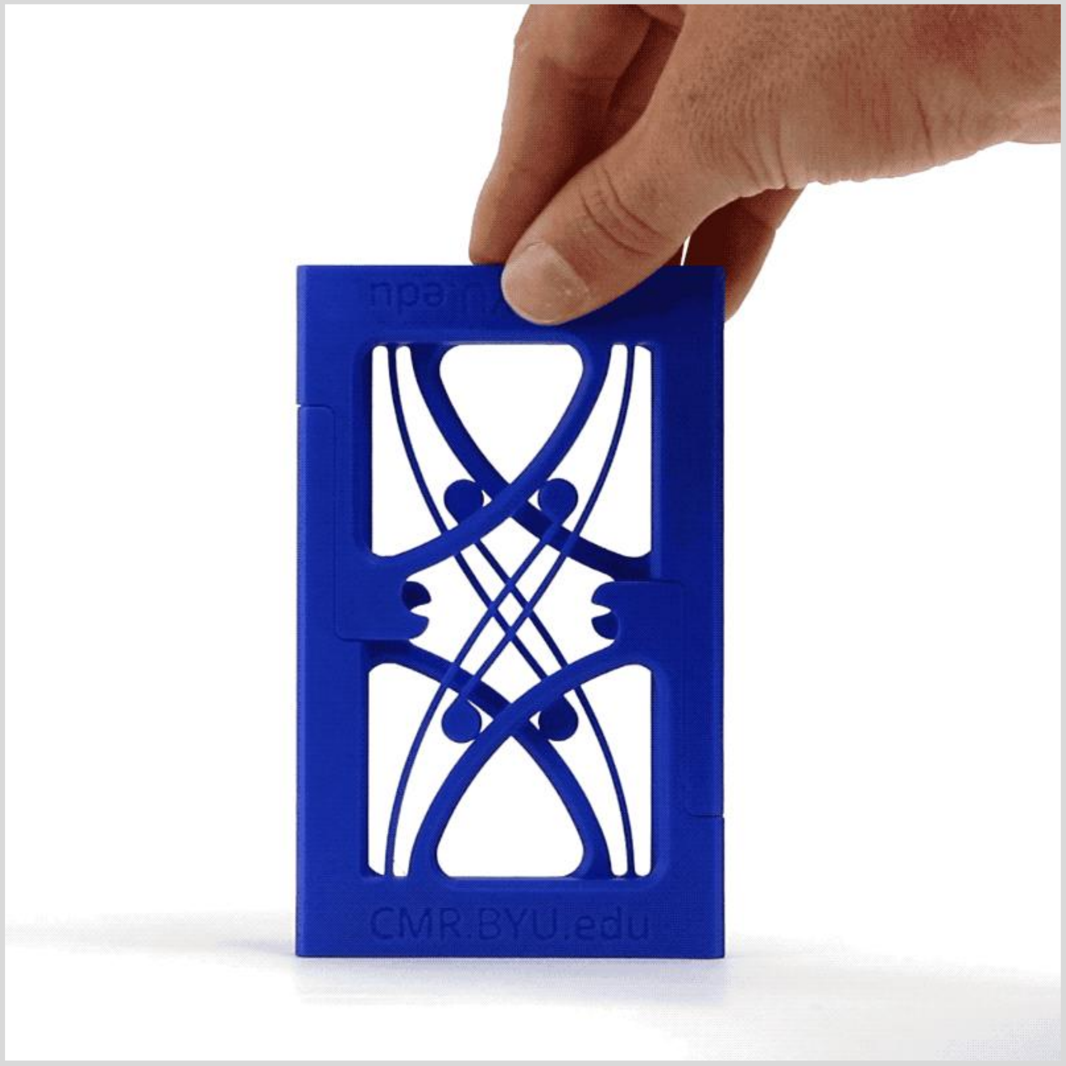
- Single print with interlocking parts.
- Assembled when the print finishes.
- *Example:* Hinges, chain links, chainmail.
- Flexify collection by printables.com/@Flying Gyroscope



Functional Flexing

- Springs and **compliant mechanisms**.
- Features/shapes that function by bending and flexing.
- Elastic materials (springy, without permanent deformation).
- Squeezing springs is literally stressful.
  - Remember strength and layer lines.
  - Flex springs in XY plane for maximum durability.

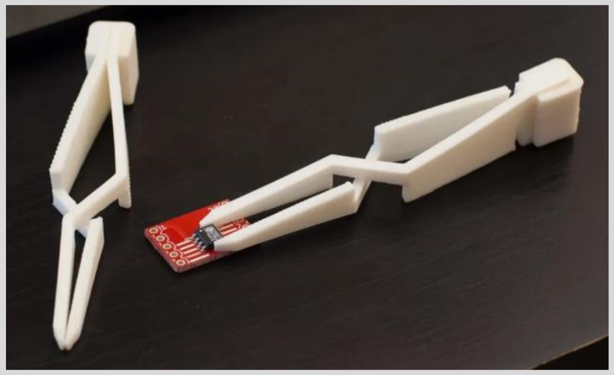
(Animation, Constant Force Mechanism by BYU CMR)



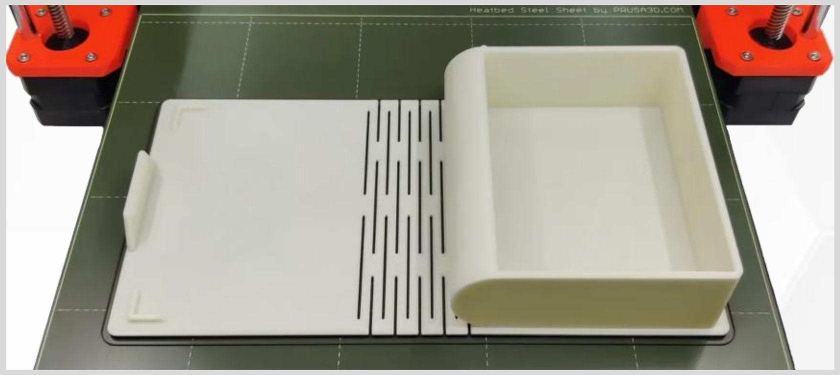
[Springs by VC Design]

[Clip by AE]

[Cross tweezer by joo]



[Living hinge by DrJones]

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